ABM BUSINESS LOGIC BIBLE

§15 v2 — RELIABILITY, RESILIENCE & ANTIFRAGILITY ENGINE

10 Iterations · 10 Layers · 27 New Rules · §15 v1's 'Complete' Verdict Revised

§15 v1 reconciled the integration-mechanics layer (retry, circuit breakers, dead-letter queues) and called §15 Complete. That verdict was right about mechanics and wrong about business logic: it conflated *the retry plumbing is correct* with *failure handling is exhausted*. §15 v2 corrects that. **§15 is not a retry-and-recovery module. It is the Reliability, Resilience and Antifragility Engine.** Recovery exists to restore reliable DECISION-MAKING, not systems — and the highest form of recovery converts a breakdown into a new organizational capability, not merely a closed incident.

Iterations	10	**Layers**	10
New rules	27	**Integration decisions**	4 explicit

FOUR INTEGRATION DECISIONS

Failure Cost / Resilience Investment reuse the Bible's existing Expected-Value family (§6 v3 marginal-value gating, §5 v3 TIER-PORT-002 ranking) — not a third value-estimation method.

Recovery Capacity IS §5 v3's Attention Economy, applied to recovery effort specifically — recovery attention is spent from the same TIER-ATTN budget as everything else scarce in this Bible, not a new currency.

Repeat-Failure escalation generalizes §8 v2/§13 v2's Failure Cluster mechanism (QC-SYS-01/02) from QC-specific clusters to ALL §15 failure types — QC-SYS-01 turns out to be an earlier, narrower instance of this more general mechanism, not a separate one.

Antifragility's Capability Improvement Proposal still requires A4 governance approval — sounding positive ('we got stronger') is not an exemption from GOV-MOD-01's human-approval requirement for any architecture change.

THE TEN LAYERS — §15 v2 ARCHITECTURE MAP

#	Layer	From iter.	What it adds
1	Failure Severity (Sev)	1	Business Severity is independent of failure TYPE — a P3 CRM sync failure and a P0 one use the same retry mechanics, different urgency
2	Cascading Failures (Cascade)	2	Most failures are dependency failures; fix the root, not each downstream symptom
3	Recovery Capacity (Cap)	3	Recovery effort is an Attention-Economy expenditure, not an assumed-infinite resource
4	Failure Cost (Cost)	4	Business Impact, not failure count, drives prioritization — reuses existing Expected-Value mechanics
5	Reliability Preservation (Trust)	5	The asset protected is organizational trust in the engine, not infrastructure uptime
6	Detection Latency (Lat)	6	MTTD tracked separately from MTTR — often the more expensive half
7	Recovery Verification (Verify)	7	A failure stays open until recovery is VERIFIED, not merely attempted
8	Repeat Failure (Repeat)	8	Generalizes §8 v2/§13 v2 Failure Clusters to every §15 failure type
9	Resilience Investment (Invest)	9	Fix by Expected Loss Prevented, not incident volume — a portfolio allocation problem
10	Antifragility (Anti)	10	The deepest discovery: recovery's highest form is capability creation, not case closure

LAYER 1 — FAILURE SEVERITY (SEV)

A missed webhook and a failed founder-escalation email are both, mechanically, notification failures — §15 v1's FAIL-* classification treats them identically. But their business consequences are nothing alike. Failure Type and Failure Impact are independent axes.

Rule ID	Phase	Specification
FAIL-SEV-01	P1	Every failure receives a Business Severity (P0 Critical / P1 High / P2 Medium / P3 Low), assigned INDEPENDENTLY of its §15 v1 failure type. The same failure type (e.g. CRM sync failure) can carry different severities depending on the specific object affected — a sync failure on a PROPOSAL-stage Opportunity is P0; the same failure type on a WATCHING account is P3.
FAIL-SEV-02	P1	Retry urgency and recovery-queue priority are driven by Business Severity, not failure type alone — a P0 transient failure and a P3 transient failure both get exponential backoff per FAIL-02, but the P0 case escalates to human attention far sooner if backoff doesn't resolve it.

LAYER 2 — CASCADING FAILURE THEORY (CASCADE)

Rule ID	Phase	Specification
FAIL-CASC-01	P1	Every failure stores its Upstream Dependency — what system or service it required to succeed. An enrichment-API outage that subsequently prevents contact discovery, which prevents outreach, which prevents engagement, which degrades scoring, is recorded as ONE root failure with a dependency chain, not four-plus unrelated incidents.
FAIL-CASC-02	P2	Failure dashboards display a Root Failure Tree (per OBS, §4 v2), not merely aggregate error counts — a spike in four different downstream error types that share one upstream dependency renders as one root-cause branch, not four alarms.
FAIL-CASC-03	P1	Recovery effort prioritizes root causes before downstream symptoms: fixing the enrichment-API outage resolves the contact-discovery, outreach, engagement, and scoring symptoms simultaneously; recovering each symptom independently while the root cause persists wastes the Layer 3 recovery-capacity budget on guaranteed recurrence.

LAYER 3 — RECOVERY CAPACITY (CAP)

Dead-letter queues are reviewed daily (§15 v1 RETRY-02) — but what if 500 items land in one day? Human recovery attention is finite, exactly like every other scarce resource this Bible tracks.

Rule ID	Phase	Specification
FAIL-CAP-01	P2	A Recovery Backlog is tracked per failure category, consumed from the SAME attention-unit currency as §5 v3 TIER-ATTN-001 (reviewer-minutes) — recovery review is not a free, infinitely-available action; it draws from the engine's overall attention budget like any other human-effort expenditure in the Bible.
FAIL-CAP-02	P1	The recovery queue is prioritized by Business Severity (Layer 1) × Opportunity Impact, NOT arrival order — the same ranking discipline as §5 v3 TIER-PORT-002 (Expected Opportunity Value ÷ Resource Cost), applied to recovery-item ordering rather than HOT-admission ordering.
FAIL-CAP-03	P2	DLQ health (backlog size, age distribution, recovery-capacity utilization) is a first-class §13 v2 operational KPI — unrecovered failures are hidden technical debt, and a growing backlog with stable daily review capacity is itself a Layer 9 resilience-investment signal (more recovery capacity is needed, or the inflow rate itself needs reducing at the root).

LAYER 4 — FAILURE COST THEORY (COST)

Rule ID	Phase	Specification
FAIL-COST-01	P1	Every P0/P1 failure receives a Business Impact estimate (opportunity delay, trust erosion, reputation risk, revenue risk) computed via the SAME Expected-Value family already used throughout the Bible (§6 v3 marginal-value estimation, §5 v3 TIER-PORT-002) — not a separate, third valuation method invented for failures specifically.
FAIL-COST-02	P1	Operational reporting (§13 v2) includes Failure Cost (estimated Business Impact, aggregated) alongside Failure Volume — a high-volume, low-cost failure category and a low-volume, high-cost one are never collapsed into one 'failure rate' number.
FAIL-COST-03	P2	Engineering and recovery-capacity prioritization is driven by Expected Business Loss, not raw error frequency — this is Layer 9's portfolio-allocation principle, foreshadowed here as it applies to day-to-day triage rather than quarterly investment review.

LAYER 5 — RELIABILITY PRESERVATION (TRUST)

The ABM engine makes implicit promises: signals will be available, scores will be accurate, handoffs will arrive, CRM will stay synchronized. Every broken promise erodes the SAME institutional trust §8 v2 already identified as the deepest asset QC protects — §15's failures are simply a different source of the same erosion.

Rule ID	Phase	Specification
FAIL-TRUST-01	**P1**	Every failure is additionally classified by Reliability Impact (Low/Medium/High/Critical) — distinct from Business Severity (Layer 1), measuring damage to confidence IN THE PLATFORM specifically (will humans keep trusting the engine's outputs), mirroring §8 v2 QC-TRUST-01's Trust Impact taxonomy applied to infrastructure failures rather than QC failures.
FAIL-TRUST-02	**P2**	Operational reviews evaluate Reliability Preservation (is institutional trust in the platform holding or eroding) alongside uptime metrics — high uptime with declining Reliability Impact scores (frequent small breakages that erode confidence even without major outages) is reported as a warning, the §15 instance of the same pattern §13 v2 ANL-KPI-04 already established for Trust Preservation Score.
FAIL-TRUST-03	**P1**	§15's own success metric is Trustworthy Operation, not Retry Success Rate — a system that retries flawlessly but whose failures still visibly erode user confidence (because they're frequent, slow to resolve, or poorly communicated) has not actually succeeded at §15's real job.

LAYER 6 — DETECTION LATENCY (LAT)

A CRM sync breaks at 09:00, is detected at 16:00, and recovers in 30 minutes by 16:30. The true outage was 7.5 hours; recovery itself took 30 minutes. Detection delay, not recovery speed, was the real cost — and §15 v1 never separately measured it.

Rule ID	Phase	Specification
FAIL-LAT-01	P1	Every major failure tracks MTTD (Mean Time To Detect — failure occurrence to detection) and MTTR (Mean Time To Recover — detection to verified resolution, per Layer 7) as SEPARATE metrics, never blended into one 'incident duration' figure.
FAIL-LAT-02	P1	Failure reviews prioritize closing detection bottlenecks BEFORE recovery bottlenecks when MTTD exceeds MTTR for a given failure category — a fast recovery process is wasted value if the failure isn't noticed for hours; FAIL-08's no-silent-failure principle (§15 v1) is necessary but not sufficient — 'logged' is not the same as 'detected promptly.'
FAIL-LAT-03	P2	Detection Lag is recorded as a first-class metric on every major (P0/P1) failure record, feeding the same §13 v2 operational dashboard as DLQ health (Layer 3).

LAYER 7 — RECOVERY VERIFICATION (VERIFY)

Rule ID	Phase	Specification
FAIL-VERIFY-01	P1	Every recovery action produces a Recovery Confidence Score — not merely a success/failure flag on the retry attempt itself. A CRM replay that executes without throwing an error has not necessarily verified that all records synced correctly, ordering was preserved (§12 v2 Layer 7), or no duplicates were introduced.
FAIL-VERIFY-02	P1	Critical (P0/P1) recoveries require explicit Post-Recovery Validation — a check that the recovered state actually matches expected reality, not merely that the recovery action completed without throwing an exception.
FAIL-VERIFY-03	P1	A failure's state remains OPEN until recovery is VERIFIED, not merely attempted — this redefines MTTR (Layer 6) to mean time-to-verified-recovery, closing a gap where a technically-executed-but-unverified recovery could be mis-reported as resolved while actually leaving corrupted or incomplete state behind.

LAYER 8 — REPEAT FAILURE THEORY (REPEAT)

A LinkedIn ban-risk circuit breaker that trips, recovers, and trips again a week later on the same pattern is not three independent incidents — it's evidence that the FAIL-LEARN-01 Failure Pattern Record from the first occurrence didn't actually prevent the second. This generalizes a mechanism §8 v2 and §13 v2 already built for QC specifically.

Rule ID	Phase	Specification
FAIL-REPEAT-01	**P1**	Every failure recurrence (matched against prior Failure Pattern Records, FAIL-LEARN-01) increments a Failure Recurrence Count for that specific failure signature, tracked across ALL §15 failure types — GENERALIZING §8 v2 QC-SYS-01 / §13 v2's Failure Cluster mechanism from QC-specific clusters to every category in §15. QC-SYS-01 is recognized as the earlier, narrower instance of this more general mechanism.
FAIL-REPEAT-02	**P1**	Recurrence automatically escalates Business Severity (Layer 1): a 3rd occurrence of the same failure signature triggers a mandatory Root Cause Review regardless of its original severity rating — a P3 failure that recurs repeatedly is no longer treated as P3, because the recurrence itself is evidence of an unaddressed systemic issue.
FAIL-REPEAT-03	**P1**	The goal of repeat-failure handling is to ELIMINATE the failure class, not merely resolve each incident — a Root Cause Review's output is judged on whether the SAME signature stops recurring afterward, not on whether the triggering incident was individually closed.

LAYER 9 — RESILIENCE INVESTMENT THEORY (INVEST)

Engineering effort naturally gravitates to the most FREQUENT failures. That's often wrong: a rare, high-impact failure (the founder-escalation email that silently fails once a quarter, costing a relationship) deserves more attention than a frequent, low-impact one. This is a portfolio allocation problem, not a frequency-sorting problem.

Rule ID	Phase	Specification
FAIL-INVEST-01	P2	A Resilience Opportunity Register tracks candidate reliability improvements (better detection, better recovery automation, root-cause elimination) as a standing backlog, parallel in structure to §6 v3's Shared Intelligence Assets pattern — a durable, queryable register, not an ad-hoc list.
FAIL-INVEST-02	P1	Resilience fixes are prioritized by Expected Loss Prevented (reusing Layer 4's Business Impact estimation method), NOT by raw incident count — a rare P0 failure with high Business Impact can outrank a frequent P3 annoyance on the Register, exactly mirroring §5 v3 TIER-PORT-002's ratio-based ranking rather than a simple frequency sort.
FAIL-INVEST-03	P2	A quarterly review examines the TOP 10 preventable failure SOURCES by Expected Loss Prevented, not the top 10 by failure VOLUME — paired with §13 v2's Learning Review Board cadence (ANL-LCHANGE-02) as a natural shared quarterly governance moment, though a distinct review with its own agenda.

LAYER 10 — ANTIFRAGILITY (ANTI)

The deepest discovery. Most systems stop at 'recovered, case closed.' The highest form of recovery converts a breakdown into a capability the organization didn't have before — better detection, a better playbook, a predictive warning that would have caught the NEXT occurrence before it happened.

Rule ID	Phase	Specification
FAIL-ANTI-01	**P1**	Every major (P0/P1, or any P2/P3 that recurs per Layer 8) incident produces a Capability Improvement Proposal — not merely a root-cause writeup. The proposal names a SPECIFIC new capability (a new detection rule, an automated pre-check, a predictive signal) that would have prevented or caught this failure class earlier.
FAIL-ANTI-02	**P1**	Failure reviews explicitly answer: 'what new capability exists because this happened?' — a review that produces only a root-cause explanation, with no proposed capability improvement, is incomplete by this standard.
FAIL-ANTI-03	**P1**	INTEGRATION DECISION: a Capability Improvement Proposal is still an architecture change and REQUIRES A4 governance approval (GOV-MOD-01) before implementation — framing it as resilience-positive does not exempt it from the same human-approval discipline every other learned change in this Bible requires (§13 v2 Loops, §6 v3 calibration, etc.).
FAIL-ANTI-04	**P2**	§15's success metric ultimately includes Organizational Resilience Growth (are Capability Improvement Proposals being generated, approved, and actually reducing Failure Recurrence Count over time, per Layer 8) — NOT incident-closure count. A team that closes many incidents but generates few capability improvements is treated as under-performing §15's real purpose, even with a clean dashboard of resolved tickets.

OPERATIONAL NOTE — THIS REQUIRES STAFFING, NOT ONLY SPECIFICATION

Several rules in this pass (FAIL-CAP, FAIL-INVEST-03, FAIL-ANTI-01/02) describe reviews and registers that require a human to actually run them on a cadence — the quarterly Resilience Investment review and the per-incident Capability Improvement review are real operational commitments, not something that executes itself once specified. This is noted explicitly so the rules aren't mistaken for self-running mechanisms; they require the same standing human governance function §13 v2's Learning Review Board already establishes a precedent for.

§15 v2 RULE ROLL-UP & PLACEMENT STATUS

Layer	Rules	Phase	Cross-section binding
1 Sev	FAIL-SEV-01..02	P1	Independent of §15 v1 failure-type classification
2 Cascade	FAIL-CASC-01..03	P1 / P2	OBS root-failure-tree dashboards
3 Cap	FAIL-CAP-01..03	P1 / P2	§5 v3 TIER-ATTN-001/TIER-PORT-002 (reused, not duplicated)
4 Cost	FAIL-COST-01..03	P1 / P2	§6 v3 marginal-value, §5 v3 TIER-PORT-002 (same Expected-Value family)
5 Trust	FAIL-TRUST-01..03	P1 / P2	§8 v2 QC-TRUST-01, §13 v2 ANL-KPI-04 (same pattern)
6 Lat	FAIL-LAT-01..03	P1 / P2	§15 v1 FAIL-08 (necessary but insufficient alone)
7 Verify	FAIL-VERIFY-01..03	P1	Redefines MTTR; §12 v2 ordering/dedup as validation targets
8 Repeat	FAIL-REPEAT-01..03	P1	Generalizes §8 v2/§13 v2 QC-SYS-01 (now its special case)
9 Invest	FAIL-INVEST-01..03	P1 / P2	§6 v3 Shared Intelligence Assets (structural pattern), §13 v2 Learning Review Board cadence
10 Anti	FAIL-ANTI-01..04	P1 / P2	GOV-MOD-01 (no exemption), §13 v2 Loop discipline

§15 v2 COMPLETE — THE 'COMPLETE' VERDICT REVISED

§15 v1's mechanics-level verdict stands: retry strategies, circuit breakers, dead-letter queues, and recovery workflows ARE correctly specified and reconciled. What v1 missed is everything ABOVE the mechanics: failure severity independent of type, cascading dependency chains, finite recovery capacity, the actual business cost of a failure, the trust being protected, detection latency as often the larger cost than recovery itself, recovery verification (not just attempt), automatic escalation on recurrence, portfolio-style resilience investment, and — the deepest discovery — that the highest form of recovery converts a breakdown into a new capability rather than simply closing a ticket. 27 new rules across 10 layers, four integration decisions binding new mechanisms to ones already built rather than duplicating them.

§15 is no longer the Bible's most over-confidently 'Complete' section. It is now, honestly, the Reliability, Resilience and Antifragility Engine — and its own success metric is no longer retry success, it's whether the organization measurably gets stronger every time something breaks.

PLACEMENT MAP UPDATE

§15 status: revised from Complete to **Complete-with-Opens** is NOT needed — all ten layers resolved to concrete rules, no genuinely undecided design questions remain. §15 is Complete at both the mechanics layer (v1) and the operational-intelligence layer (v2) now.

ON THE HORIZON

§17 KSA-Specific Business Rules remains the Bible's only genuinely 'Not started' section — and per both §15 reviewers' independent assessment, likely the richest remaining territory, because it's where every universal rule built since §1 collides with real-world Saudi banking regulation, buying culture, SAMA expectations, PDPL, Ramadan timing, and Vision 2030 alignment. §16 Timing Matrix consolidation is the natural lighter companion.